

Device Needed

Device	Description	Quantity	Total Cost (RM)
Router	Cisco ISR 4000 Series, Dual WAN Ports, IPv6 support, Advanced QoS, Secure Application Hosting.	2	20000
Switch	Cisco Catalyst 9300, Layer 3, 24 Ports, Stackable, Supports VLAN, QoS, and PoE+, 10G uplink-ready.	5	25000
Patch Panel	24-Port Cat6a, Gold-plated contacts, Toolless termination, Detachable cable management bar.	8	1600
Cat6a Cables	Ethernet Cat6a, 10Gbps, 550MHz, Snagless Design, 1524cm per roll.	100 rolls	15000
Fiber Optic Cable	SC-UPC APC, Single Mode, 9/125um, Insertion Loss: 0.15dB, 1310/1550nm, ISO9001 Certified.	300m	363
Wireless Access Point	TP-Link EAP670, Wi-Fi 6, Dual-Band 5400Mbps, 204m ²	14	9212

	Coverage, Ceiling/Wall Mount, Omada SDN integration.		
Server	Cisco UCS C480M5, Dual Intel Xeon Scalable CPUs, 384GB DDR4, 32x 2.5" SAS/SATA, 10G network adapters, PCIe Gen 3.	3	60000
Firewall	FS NSG-5100, 6Gbps Throughput, Threat Detection, 2M Concurrent Sessions, 6 RJ45 Ports, IPS, QoS, 2-Year Warranty.	1	36811
PC (General Labs)	Intel Core i5, 16GB RAM, 512GB SSD, Windows 11 Pro, Gigabit Ethernet, Wi-Fi 6.	60	210000
PC (Cisco Lab)	Intel Core i7, 16GB RAM, 512GB SSD, Windows 11 Pro, Advanced network simulation compatibility.	30	135000
PC (Embedded Lab)	Intel Core i7, 32GB RAM, 1TB SSD, Windows 11 Pro, IoT and sensor simulation ready.	30	180000
Smart TV	Samsung QN900B, 75", 7680x4320 Resolution, QLED Display, 90W Sound Output, HDMI x4, Ethernet, Dolby Atmos, 120Hz Refresh Rate.	2	88252

Projector	Epson VS250 SVGA, 3200 Lumens Brightness, 800x600 Resolution, HDMI Input, Compact Design.	3	7518
UPS	10kVA, Pure Sine Wave, Battery Backup for up to 1 Hour, Network Monitoring.	2	20000
Cooling System	Precision Air Cooling System, 15kW Capacity, Dual Compressor, Temperature Monitoring, Underfloor Distribution.	2	100000
Storage (NAS/SAN)	32TB Capacity, Redundant Power Supply, Gigabit Ethernet, Backup Automation.	1	15000

Device required in each facility

Facility	Devices Required	Quantity
General Labs (2)	PCs: Intel Core i5, 16GB RAM, 512GB SSD	60 (30 per lab)
	Switches: Cisco Catalyst 9200	2
	Wireless Access Points: TP-Link EAP670	2
	Cat6a Cables	50 rolls
	Patch Panels (24-Port Cat6a)	2
Cisco Networking Lab	PCs: Intel Core i7, 16GB RAM	30
	Layer 3 Switches: Cisco Catalyst 9300	2
	Specialized Routers (Cisco ISR 4000)	1
	Next-Generation Firewall (FS NSG-5100)	1
	Wireless Access Points: TP-Link EAP670	2
	Cat6a and Fiber Optic Cables (for critical setups)	50 rolls Cat6a, 200m Fiber Optic
	Patch Panels (24-Port Cat6a)	2
Embedded Systems Lab	PCs: Intel Core i7, 32GB RAM, 1TB SSD	30
	Switches: Cisco Catalyst 9300	1
	Wireless Access Points: TP-Link EAP670	2
	Cat6a Cables	30 rolls
	Patch Panels (24-Port Cat6a)	1
Video Conferencing Room	Smart TV: Samsung QN900B	1
	Projector: Epson VS250 SVGA	1
	Wireless Access Points: TP-Link EAP670	1
	Switch	1
Hybrid Classroom	PCs: Intel Core i5, 16GB RAM, 512GB SSD	10
	Smart TV: Samsung QN900B	1
	Projector: Epson VS250 SVGA	1
	Wireless Access Points: TP-Link EAP670	1

	Switch	1
Student Lounge	Wireless Access Points: TP-Link EAP670	4
	Switch	1
	Cat6a Cables	20 rolls
Server Room	High-Performance Server: Cisco UCS C480M5	3
	Cooling Systems	1
	Uninterruptible Power Supplies (UPS)	2
Database Room	Server Backup Storage (NAS/SAN)	1
	Cooling Systems	1
Meeting Room	Wireless Access Points: TP-Link EAP670	1
	Projector: Epson VS250 SVGA	1
Others (Infrastructure)	Fiber Optic Cable	100m

Function of each device.

Device	Function	Connection Description
Router	Connects internal network to the internet and routes data between networks.	Connects to the ISP modem for internet access and to core switches for LAN connectivity.
Switch	Distributes network traffic within the LAN by connecting multiple devices.	Connects to the router for network access and to PCs, servers, and wireless access points via Ethernet cables.
Patch Panel	Organizes and manages Ethernet cables for easy connection and troubleshooting.	Connects Ethernet cables from switches, servers, and end-user devices into a centralized hub.
Cat6a Cables	Provides high-speed wired connectivity for devices within the LAN.	Connects devices like switches, PCs, patch panels, and wireless access points to ensure stable data transmission.

Fiber Optic Cable	Ensures high-speed, low-latency connections for critical network infrastructure.	Connects the main switch to the server room or database room over longer distances.
Wireless Access Point	Provides Wi-Fi connectivity for wireless devices such as laptops, smartphones, and IoT devices.	Connects to switches via Ethernet cables and extends the network wirelessly to devices.
Server	Hosts applications, stores data, and manages network resources.	Connects to switches via Ethernet for data communication with PCs and other devices.
Firewall	Protects the network from unauthorized access, viruses, and other threats.	Positioned between the router and the internal network to monitor and filter incoming and outgoing traffic.

Smart TV	Displays content for video conferencing and teaching in high resolution.	Connects wirelessly or via Ethernet to access video conferencing platforms and shared content from the server.
Projector	Displays presentations and teaching materials for classrooms and meeting rooms.	Connects to PCs or servers via HDMI for multimedia output.
UPS	Provides backup power to critical devices during outages.	Connects directly to servers, switches, and other critical infrastructure for power continuity.
Cooling System	Maintains optimal temperature for network devices in server and database rooms.	Operates independently but ensures the environment is safe for devices connected in the network.
Storage (NAS/SAN)	Centralized storage for data backups and file sharing.	Connects to the server and switches via Ethernet to provide accessible and redundant storage for network users.